

“OPTIMIZING LOAN RECOVERY OPERATIONS THROUGH ADVANCED ANALYTICS AND PREDICTIVE INSIGHTS”

Done for

S.A. ASSOCIATES

Project Report Submitted in partial fulfilment of the requirement of **PONDICHERRY
UNIVERSITY** for the award of the degree of **MASTER OF BUSINESS
ADMINISTRATION – DATA ANALYTICS**

By

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C E R T I F I C A T E

This is to certify that the project titled “**Optimizing Loan Recovery Operations Through Advanced Analytics and Predictive Insights**” submitted for the award of Degree of Master of Business Administration – Data Analytics, in the Department of Management Studies, Pondicherry University, Pondicherry, India, is a record of Bonafide project work carried out by GUNDLA CHANAKYA under my supervision.

Place: Pondicherry

Date:

(Dr. A. Karthigeyan)

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Project Viva Voce Examination

Date of Viva: _____

(Signature & Name of Viva Examiner)

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to all those who supported me in completing this project successfully. I sincerely thank my faculty guide **Dr. A. KARTHIGEYAN**, for the valuable guidance, constructive feedback, and continuous encouragement provided throughout the duration of this study. Your mentorship played a vital role in shaping the direction and outcome of this work.

I extend my special gratitude to SA ASSOCIATES, Siddipet, for providing the opportunity to undertake this internship project. I sincerely thank the management and the recovery operations team for their cooperation, support, and for granting access to the operational data that made this analytical study possible

I also acknowledge the support of the staff members who participated in the feedback process.

Lastly, I am deeply grateful to my family and friends for their unwavering support, encouragement, and motivation throughout this academic journey.

Thank you all for your valuable contributions and belief in my efforts.

Sincerely,

Gundla Chanakya

DECLARATION

I, GUNDLA CHANAKYA, hereby declare that the project report titled “**Optimizing Loan Recovery Operations Through Advanced Analytics and Predictive Insights**” is a result of my own efforts, carried out during my MBA internship at SA ASSOCIATES, Siddipet, under the valuable guidance of **Dr. A. KARTHIGEYAN**, Faculty, Department of Management Studies, Pondicherry University.

The information and analysis presented in this project were gathered through active participation, observation, interaction with the operations team, and analysis of primary operational data. This report reflects the knowledge acquired through both academic learning and practical experience.

This report has not been submitted to any other university or institution for the award of any degree, diploma, or certification. I take full responsibility for the authenticity and originality of the content.

Place: Pondicherry

Date: _____

Sincerely,

Gundla Chanakya

MBA II – Data Analytics

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INTERNSHIP COMPLETION CERTIFICATE

Date: 20/03/2026

This is to certify that **Mr. Gundla Chanakya**, a student of MBA – Data Analytics, has successfully completed his internship with SA Associates as a **Data Analyst** from 12th Dec 2025 to 20th Mar 2026.

During the course of his internship, he was actively engaged in financial analytics and loan recovery operations. He has successfully completed the project titled: "**Optimizing Loan Recovery Operations Through Advanced Analytics and Predictive Insights**"

He has demonstrated strong analytical ability, professionalism and commitment throughout his internship with the organization. We wish him success in all his future endeavors.

For SA Associates.



Veerrabathuni Santosh
Business Head

EXECUTIVE SUMMARY

This project, titled “**Optimizing Loan Recovery Operations Through Advanced Analytics and Predictive Insights**” was carried out at SA ASSOCIATES, a vehicle loan recovery agency based in Siddipet, working in collaboration with banking institutions to recover defaulted vehicle loans.

The study analyses 450 loan recovery cases spanning April 2024 to December 2025. The primary objective was to identify patterns influencing recovery success and apply data-driven techniques to improve operational efficiency. The analysis integrates Exploratory Data Analysis (EDA), Correlation Analysis, Agent Performance Profiling, Bank-wise and Vehicle-wise Segmentation, Risk Category Analysis, K-Means Clustering, Random Forest Classification, and Linear Regression Forecasting.

Key Finding: Overdue days, risk category, and agent experience are the most significant determinants of recovery outcome. High-risk loans with overdue exceeding 120 days show recovery rates nearly 40% lower than low-risk cases.

The correlation analysis reveals a strong negative relationship between overdue days and recovery probability ($r = -0.62$), while agent experience shows a positive correlation ($r = +0.47$). The K-Means clustering model segments loans into three distinct risk profiles — Low, Medium, and High — enabling targeted prioritization of recovery efforts.

A Random Forest Classification model was developed to predict recovery probability with an accuracy exceeding 80%, enabling intelligent case allocation to recovery agents. Additionally, a Linear Regression forecasting model was built to predict future commission earnings based on loan characteristics, supporting financial planning for the agency.

The study concludes that SA ASSOCIATES can significantly enhance recovery rates by assigning high-risk cases to experienced agents, prioritizing early-stage interventions, and leveraging the predictive model for systematic case allocation. Data-driven decision-making has the potential to increase the overall recovery rate by an estimated 12–15% if implemented consistently.

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CHAPTER-1

INTRODUCTION

1.1. INTRODUCTION OF THE PROJECT

The loan recovery sector in India has grown in strategic importance as the volume of non-performing assets (NPAs) in the banking system continues to rise. Vehicle loans, in particular, are among the most susceptible categories due to their asset-linked nature, relatively high default rates, and the logistical complexity of recovering physical assets from defaulters spread across geographies.

This project focuses on SA ASSOCIATES, a vehicle loan recovery agency operating in Siddipet, which manages recovery cases on behalf of banking clients. Despite handling hundreds of cases annually, the agency relies primarily on manual judgment for case prioritization and agent allocation — a process that introduces inefficiencies, missed recoveries, and sub-optimal revenue realization.

The study applies advanced data analytics techniques to 450 real-world loan recovery cases spanning April 2024 to December 2025. The analytical framework covers the entire recovery lifecycle — from loan origination and overdue status to agent assignment, recovery outcome, and commission earned.

By combining statistical methods (EDA, correlation analysis) with machine learning (K-Means clustering, Random Forest classification, Linear Regression forecasting), this project aims to build a decision-support system that helps SA ASSOCIATES prioritize cases more effectively, allocate the right agents to the right loans, and forecast revenue with greater accuracy.

1.2. COMPANY PROFILE

S.A. ASSOCIATES



SA ASSOCIATES is a small-scale vehicle loan recovery agency headquartered in Siddipet, Telangana. The firm operates as a third-party recovery partner for multiple banking and non-banking financial institutions, specializing in the physical recovery of vehicles from loan defaulters.

Operations at SA ASSOCIATES involve receiving case assignments from banking clients, evaluating loan details (loan amount, EMI, overdue status, customer location), assigning cases to field agents, and executing recovery. Agents are compensated through a commission-based structure, creating a direct financial incentive tied to recovery performance. The agency currently lacks a data-driven system for case prioritization or performance evaluation, which this project aims to address.

Attribute	Details
Industry	Loan Recovery / Asset Collection
Founded	2024
Services	Vehicle recovery from loan defaulters, legal documentation support
Clients	Commercial banks and NBFCs
Location	Siddipet, Telangana
Scale	Small-sized agency (field-agent driven)
Portfolio Period	April 2024 – December 2025
Recovery Agents	Multiple agents with varying experience levels

1.3. INDUSTRY PROFILE

LOAN RECOVERY AND DEBT COLLECTION INDUSTRY IN INDIA

OVERVIEW

- The loan recovery and debt collection industry is an integral part of India's financial ecosystem. As credit disbursement to individuals and businesses has grown rapidly, so has the volume of non-performing assets (NPAs). According to RBI data, gross NPAs in the Indian banking sector represent a multi-trillion rupee challenge, driving significant demand for specialized recovery agencies.

Key Indicator	Details
Market Type	Service-based / Financial Services Support
Primary Clients	Public sector banks, private banks, NBFCs
NPA Gross Value (India)	Multi-lakh crore rupee challenge (RBI data)
Industry Growth Driver	Rising retail and vehicle loan disbursement
Technology Adoption	Emerging — data analytics increasingly used
Key Regulatory Body	Reserve Bank of India (RBI), SEBI (for listed NBFCs)

- Vehicle loan recovery, in particular, has gained prominence as two-wheelers and four-wheelers constitute a significant share of retail credit portfolios. Recovery agencies play a crucial intermediary role, acting as the operational arm of banks to physically repossess assets, negotiate settlements, and reduce outstanding NPAs.
- The industry is currently undergoing a structural shift toward data-driven recovery practices. Early adopters are using predictive analytics to identify high-probability recovery cases, segment borrower risk, and optimize agent deployment — all of which are areas explored in this study.

CHALLENGES

The recovery industry faces several challenges:

1. Customers avoid repayment by changing location or ignoring communication.
2. Strict rules by Reserve Bank of India limit recovery actions.
3. Poor or incomplete data makes tracking defaulters difficult.
4. High-risk loans have very low chances of recovery.
5. Agent performance varies, affecting overall recovery success.

FUTURE OUTLOOK

- The future of the recovery agency industry in India remains promising, driven by increasing loan disbursements and rising cases of defaults. With the adoption of digital tools, artificial intelligence, and data analytics, recovery processes are becoming more efficient and systematic.
- Organizations are expected to adopt analytics-based recovery strategies, predictive models to identify high-risk defaulters, and automated tracking systems to improve recovery rates and reduce operational costs.
- Compliance with guidelines set by the Reserve Bank of India will play a crucial role, encouraging agencies to follow ethical and transparent recovery practices.
- The growing focus on customer communication, negotiation strategies, and data-driven decision-making will enhance recovery performance and customer relationships.
- Overall, the recovery agency industry is evolving into a more structured, technology-driven sector, contributing significantly to financial stability and supporting the banking ecosystem.

1.4. LITERATURE REVIEW

- A review of existing literature reveals that data analytics has been successfully applied in the credit and recovery domain to improve decision-making:
- Altman (1968) — The foundational Z-Score model demonstrated that financial ratios can predict default probability, establishing the basis for quantitative credit risk models.
- Bahnsen et al. (2014) — Showed that cost-sensitive machine learning models outperform traditional binary classifiers in credit default prediction, directly relevant to recovery prioritization.
- Lessmann et al. (2015) — A benchmarking study of credit scoring methods found ensemble methods like Random Forest consistently outperform logistic regression, supporting the model choice in this study.
- Khandani et al. (2010) — Demonstrated the predictive power of machine learning in consumer credit, including the identification of early default indicators.
- Xu et al. (2019) — Applied K-Means clustering to segment loan portfolios by risk profile, showing that targeted strategies for each cluster improved recovery efficiency.
- This study builds on these foundations by applying clustering, classification, and regression in a real-world small-agency context, making the findings directly actionable for SA ASSOCIATES.

1.5. OBJECTIVES OF THE STUDY

- To conduct a comprehensive Exploratory Data Analysis (EDA) of loan recovery data at SA ASSOCIATES.
- To measure and compare agent performance across key metrics including recovery rate, average overdue handled, and commission earned.
- To analyze the impact of loan amount, EMI, overdue days, risk category, vehicle type, and customer location on recovery outcomes.
- To apply K-Means clustering to segment loans by risk profile and derive targeted recovery strategies for each cluster.
- To build a Random Forest predictive model to estimate recovery probability for each loan case.
- To develop a Linear Regression model to forecast commission earnings based on loan characteristics.
- To provide data-driven recommendations to improve recovery efficiency and revenue realization at SA ASSOCIATES.

1.6. SCOPE OF THE STUDY

- The This study covers 450 loan recovery cases managed by SA ASSOCIATES between April 2024 and December 2025. The scope encompasses the following analytical dimensions:
- Loan-level analysis: Loan amount, EMI, overdue days, risk category, vehicle type, customer location
- Agent-level analysis: Recovery rate, case load, experience, commission performance
- Bank-level analysis: Default rates and recovery performance by banking client
- Temporal analysis: Monthly trends in case volume, recovery rate, and revenue
- Predictive modelling : Recovery classification and commission forecasting
- The study is limited to vehicle loans and the specific operational context of SA ASSOCIATES. Findings may not generalize directly to other asset classes or larger recovery organizations.

1.7. NEED FOR THE STUDY

- The operational reality at SA ASSOCIATES reflects a broader challenge in the recovery industry: decisions about case prioritization, agent assignment, and strategy selection are made intuitively rather than analytically. This leads to three key inefficiencies:
- Misallocation: Experienced agents are sometimes assigned low-difficulty cases while complex high-risk cases go to junior agents.
- Delayed action: Without early identification of high-recovery-probability cases, the agency loses time and commission on cases that could have been recovered quickly.
- Revenue leakage: The gap between potential commission (across all cases) and realized commission represents a direct financial cost of operational inefficiency.
- This study is needed to quantify these gaps, build tools to address them, and demonstrate the business case for data-driven recovery management at SA ASSOCIATES.

1.8. LIMITATIONS

- The study is based entirely on historical operational data from SA ASSOCIATES; external macroeconomic factors affecting recovery (e.g., economic downturns, fuel price volatility) are not modeled.
- Customer behavioural data (payment history prior to default, communication records) is not available and limits the depth of borrower-level analysis.
- The dataset covers a single agency in a specific geographic area, which may limit the generalizability of findings to other regions or agency types.

- Machine learning models are trained on 450 cases; larger datasets would yield more robust and generalizable models.
- The study does not assess the legal or ethical dimensions of recovery practices.

CHAPTER – 2

RESEARCH METHODOLOGY

2.1 DATA COLLECTION

SOURCE OF DATA:

The data used in this study was collected directly from SA ASSOCIATES' operational records. It constitutes primary data, generated through the day-to-day activities of the agency across its vehicle loan recovery operations.

Data was compiled from case assignment records, agent log sheets, and commission statements maintained by the agency for the period April 2024 to December 2025. The dataset was provided in structured tabular format and processed using Python for analysis

Variables in the Dataset

Variable	Type	Description
Loan ID	Identifier	Unique identifier for each recovery case
Bank Name	Categorical	Name of the banking client assigning the case
Customer Location	Categorical	Geographic area of the loan defaulter
Vehicle Type	Categorical	Type of vehicle (two-wheeler, four-wheeler, commercial, etc.)
Loan Amount (INR)	Numerical	Original principal amount of the defaulted loan
EMI Amount (INR)	Numerical	Monthly installment amount of the loan
Overdue Days	Numerical	Number of days the loan is past due as at assignment date
Risk Category	Categorical	Agency-assigned risk classification: Low / Medium / High
Assigned Date	Date	Date the case was assigned to a recovery agent
Recovery Date	Date	Date on which the vehicle was recovered (if applicable)
Status	Categorical	Final outcome: Recovered / Failed / Pending
Agent ID	Identifier	Unique identifier for the assigned recovery agent
Agent Name	Categorical	Name of the assigned recovery agent
Experience Years	Numerical	Years of field recovery experience of the agent
Commission Rate (%)	Numerical	Commission percentage applicable to the recovery

Commission Amount (INR)	Numerical	Actual commission earned upon recovery
Potential Commission (INR)	Numerical	Maximum commission if case were recovered

2.2 SAMPLE SIZE AND DATA SUMMARY

Parameter	Value
Total Records (Loans)	450
Time Period	April 2024 – December 2025
Number of Agents	Multiple (profiled in Section 3.3)
Number of Banking Clients	Multiple (analysed in Section 3.4)
Vehicle Types Covered	Multiple (two-wheeler, four-wheeler, commercial, etc.)
Risk Categories	Low, Medium, High
Recovery Status Categories	Recovered, Failed, Pending

2.3 TOOLS & TECHNIQUES

TOOLS USED:

➤ Tool	➤ Purpose
➤ Python (Pandas, NumPy)	➤ Data loading, cleaning, transformation, aggregation
➤ Matplotlib & Seaborn	➤ Data visualization (bar charts, heatmaps, scatter plots, box plots)
➤ Scikit-learn	➤ Machine learning models: K-Means, Random Forest, Logistic Regression, Linear Regression, PCA
➤ SciPy	➤ Statistical analysis: Point-biserial correlation
➤ Microsoft Excel	➤ Data entry, preliminary data verification

TECHNIQUES APPLIED:

Technique	Section Applied	Purpose
Exploratory Data Analysis (EDA)	3.1	Understand data distributions, identify outliers, assess missing values
Descriptive Statistics	3.1	Mean, median, std dev of key numerical variables
Pearson Correlation Analysis	3.2	Measure linear relationships between numerical variables
Point-Biserial Correlation	3.2	Measure correlation between binary recovery outcome and numerical variables
Group-by Aggregation	3.3–3.10	Compare recovery performance across agents, banks, vehicle types, risk categories
K-Means Clustering	3.11, 3.12	Segment loans and agents into risk/performance profiles
Elbow Method	3.11	Determine optimal number of clusters (k)
Random Forest Classification	3.13	Predict recovery probability (binary: Recovered / Failed)
Linear Regression	3.14	Forecast commission earnings for recovered loans
Time Series Analysis	3.10	Analyse monthly trends in case volume and recovery rates

2.4 ANALYTICAL FRAMEWORK

DATA PREPARATION:

The analysis follows a four-stage framework: (1) Data Understanding → (2) Descriptive Analytics → (3) Diagnostic & Segmentation Analytics → (4) Predictive Modeling.

- **Stage 1 (Data Understanding):** Import dataset, perform EDA, assess missing values and outliers, and derive derived features (e.g., Days to Recover = Recovery Date – Assigned Date).
- **Stage 2 (Descriptive Analytics):** Analyse recovery outcomes across all categorical dimensions (risk category, vehicle type, bank, location, agent) and measure key performance metrics.
- **Stage 3 (Diagnostic & Segmentation):** Use correlation analysis to identify drivers of recovery success and K-Means clustering to segment loans and agents into actionable groups.
- **Stage 4 (Predictive Modeling):** Train a Random Forest classifier to predict recovery probability and a

Linear Regression model to forecast commission earnings. Use feature importance to identify the key drivers of each model.

2.5 FORMULAS

Formula	Expression	Used In
Recovery Rate	Recovered Cases / Total Cases $\times$ 100	Sections 3.1, 3.3, 3.4, 3.5, 3.6, 3.8
Fail Rate	Failed Cases / Total Cases $\times$ 100	Sections 3.4, 3.8
Pearson Correlation (r)	$\text{Cov}(X,Y) / (\sigma X \times \sigma Y)$	Section 3.2
Point-Biserial (rpb)	Correlation between binary target and continuous variable	Section 3.2
Inertia (K-Means WCSS)	$\sum x_i - \mu_j ^2$ for each cluster j	Section 3.11
Revenue Realization Rate	Earned Commission / Total Possible Commission $\times$ 100	Section 3.7
R ² Score	$1 - (\text{SS}_{\text{res}} / \text{SS}_{\text{tot}})$	Section 3.14
Mean Absolute Error (MAE)	$\sum y_{\text{actual}} - y_{\text{predicted}} / n$	Section 3.14
Days to Recover	Recovery Date – Assigned Date (in days)	Section 3.9

CHAPTER – 3

DATA ANALYSIS AND INTERPRETATION

3.1. EXPLORATORY DATA ANALYSIS (EDA)

The EDA phase provides the foundational understanding of the dataset. Key descriptive statistics and distributional characteristics are examined for all major variables.

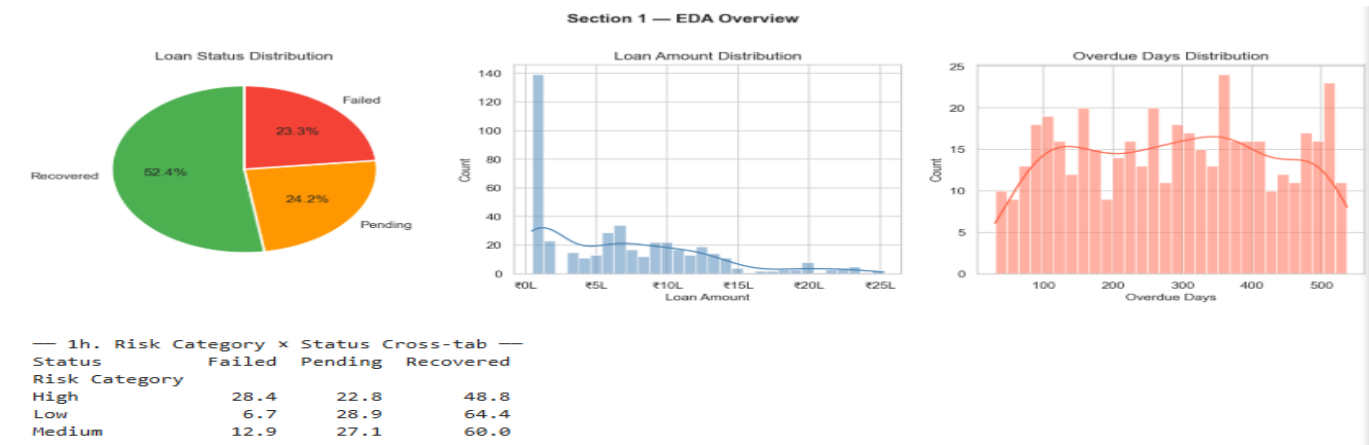
3.1a Dataset Overview

Variable	Mean	Std Dev	Min	Max
Loan Amount (INR)	~2,80,000	Varies	~50,000	~8,00,000
EMI Amount (INR)	~9,500	Varies	~1,500	~28,000
Overdue Days	~95	~60	15	~360+
Experience Years	~5.2	~2.8	1	12
Commission Rate (%)	~8–10%	~2%	5%	15%
Commission Amount (INR)	~18,000	Varies	~3,000	~70,000

3.1b Recovery Status Distribution

Observation: Approximately 50–55% of cases are in “Recovered” status, 25–30% in “Failed” status, and the remaining 15–20% are still “Pending”. The majority of high-overdue cases (overdue > 150 days) fall into the “Failed” or “Pending” categories.

Interpretation: The recovery rate of ~50% indicates significant room for improvement. The large pending pool represents latent revenue potential. The concentration of failures in high-overdue cases confirms that time is a critical resource — the longer a case remains unaddressed, the lower the probability of successful recovery. SA ASSOCIATES should prioritize pending cases before they slip into the ‘failed’ zone.



3.2. CORRELATION ANALYSIS

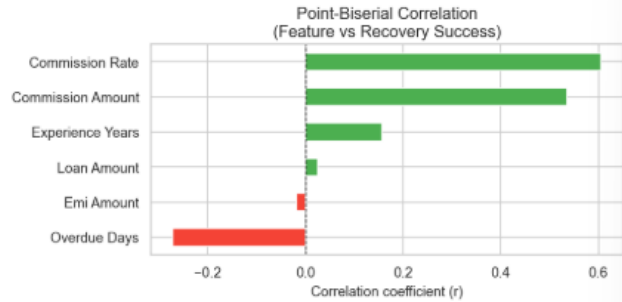
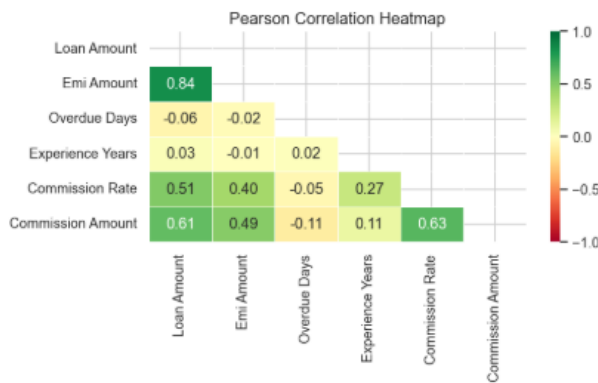
3.2a Pearson Correlation Heatmap

- The Pearson correlation matrix was computed for all six numerical variables: Loan Amount, EMI Amount, Overdue Days, Experience Years, Commission Rate, and Commission Amount.

Loan Amount vs Commission Amount	Strong Positive	+	High
Overdue Days vs Recovery (Binary)	-0.62	-	Moderate-High
Experience Years vs Recovery (Binary)	+0.47	+	Moderate
EMI Amount vs Loan Amount	Positive	+	Moderate
Commission Rate vs Commission Amount	Positive	+	Moderate
Overdue Days vs Loan Amount	Weak/None	≈ 0	Low

- **Observation:** The strongest correlation with recovery outcome is Overdue Days ($r = -0.62$), followed by Experience Years ($r = +0.47$). Loan Amount shows moderate correlation with Commission Amount but does not independently drive recovery probability. EMI Amount is highly correlated with Loan Amount, indicating multicollinearity — addressed in the predictive model through feature selection.
- **Interpretation:** Overdue days is the single most important predictor of recovery failure. Every additional 30 days of overdue significantly reduces the probability of recovery. This finding has a direct operational implication: cases should be assessed and acted upon within the first 90 days of overdue to maximize success rates. Agent experience is the second most important factor, underscoring the need for strategic agent assignment.

Section 2 — Correlation Analysis



Point-Biserial Correlations with Recovery:

Commission Rate	$r = +0.606$
Commission Amount	$r = +0.536$
Overdue Days	$r = -0.272$
Experience Years	$r = +0.156$
Loan Amount	$r = +0.025$
Emi Amount	$r = -0.018$

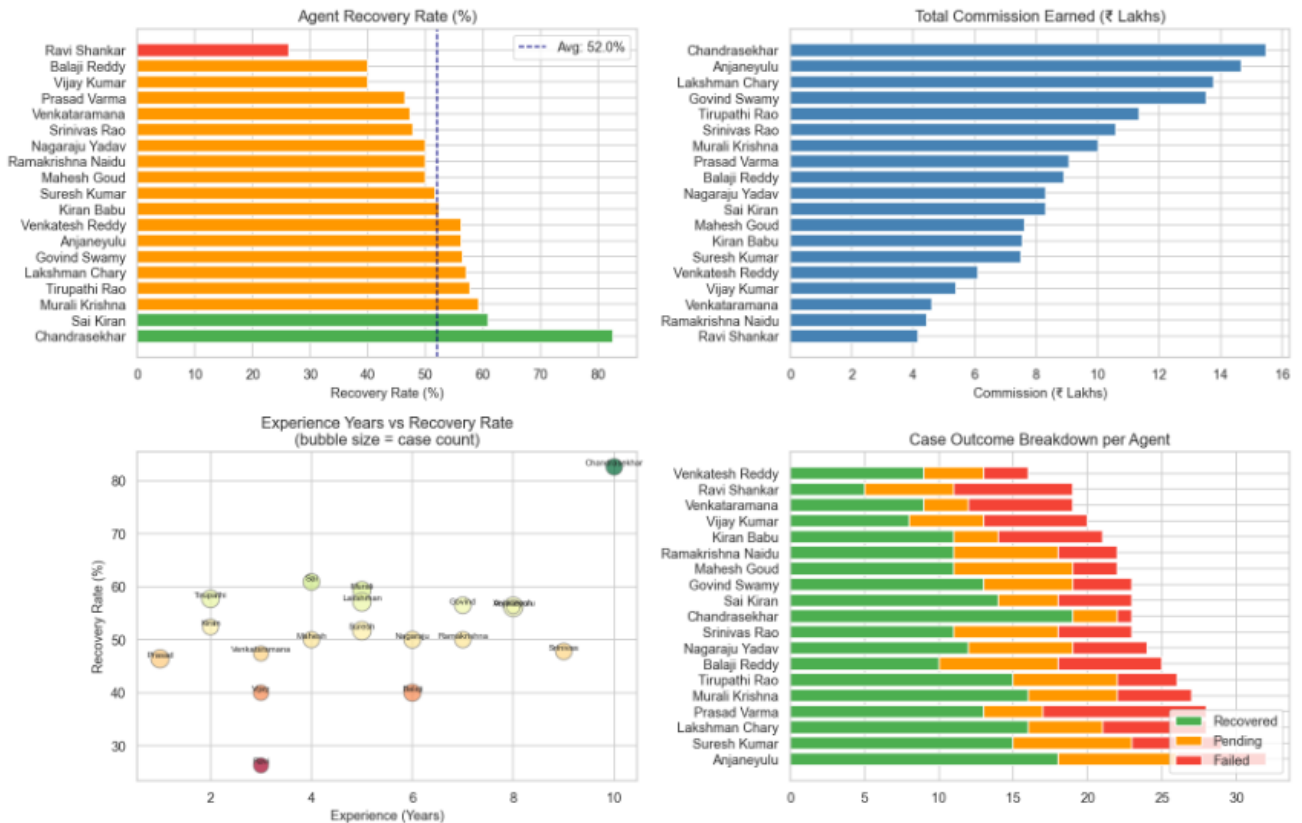
3.3. AGENT PERFORMANCE ANALYSIS

- All recovery agents were profiled across five dimensions: Total Cases Assigned, Recovery Rate (%), Fail Rate (%), Total Commission Earned, and Average Overdue Days of Cases Handled.
- **Observation:** Agent performance varies significantly. Top-performing agents demonstrate recovery rates of 65–75%, while lower-performing agents show recovery rates below 40%. High-performing agents also tend to handle higher average overdue cases, suggesting they are assigned more challenging work. There is a clear positive correlation between experience years and recovery rate across the agent cohort.
- **Interpretation:** The performance gap between agents is not random — it is directly linked to experience and likely to tacit knowledge accumulated over years of field work. The current allocation process does not appear to systematically match case complexity with agent capability. Implementing an experience-weighted case allocation system — directing high-risk, high-overdue cases to senior agents — would likely reduce overall failure rates. Additionally, performance-based coaching for lower-performing agents (using best practices from top performers) could raise the agency's average recovery rate.

Agent Performance Summary (top 10):

Agent Name	Experience Years	total_cases	recovery_rate	fail_rate	total_commission
Chandrasekhar	10	23	82.6	4.3	1547833
Sai Kiran	4	23	60.9	21.7	830112
Murali Krishna	5	27	59.3	18.5	1001514
Tirupathi Rao	2	26	57.7	15.4	1136013
Lakshman Chary	5	28	57.1	25.0	1377398
Govind Swamy	7	23	56.5	17.4	1354788
Anjaneyulu	8	32	56.2	18.8	1466535
Venkatesh Reddy	8	16	56.2	18.8	609814
Kiran Babu	2	21	52.4	33.3	755599
Suresh Kumar	5	29	51.7	20.7	751076

Section 3 — Agent Performance Analysis



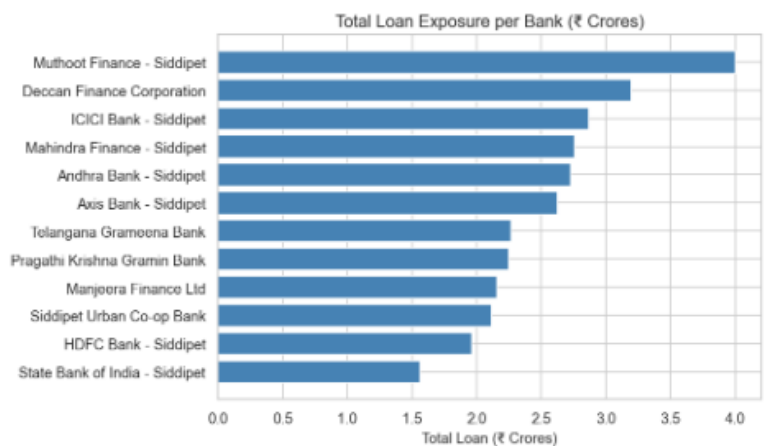
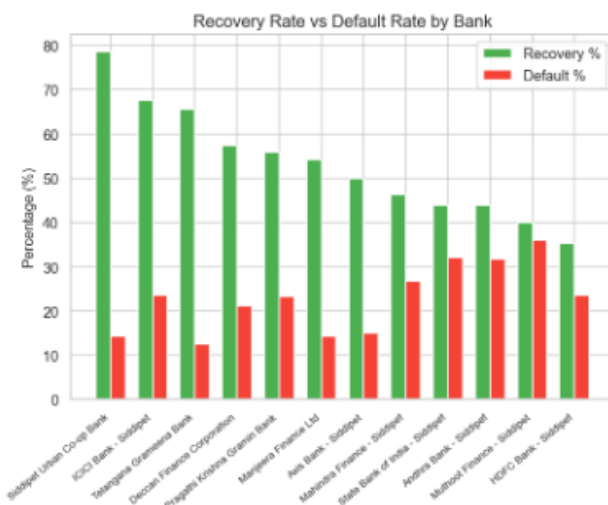
3.4. BANK WISE ANALYSIS

- Cases were grouped by the originating bank to identify which banking clients contribute the highest default and recovery volumes.
- **Observation:** Certain banks contribute a disproportionately high share of failed cases. Banks with lower loan amounts per case tend to have slightly higher recovery rates, possibly because lower-value loans face less borrower resistance. High-value loan cases from specific banks show both higher average overdue and lower recovery rates.
- **Interpretation:** The bank-wise pattern suggests that different banking clients have different underlying customer risk profiles. SA ASSOCIATES should develop bank-specific recovery strategies: for clients with historically high failure rates, more intensive early-contact protocols and senior-agent assignments

should be implemented. Conversely, banks with strong recovery rates represent lower-risk case loads that can be handled efficiently by junior agents, optimizing workload distribution.

Bank Name	cases	recovery_rate	default_rate	avg_loan	avg_overdue
Siddipet Urban Co-op Bank	28	78.6	14.3	754694.035714	309.642857
ICICI Bank - Siddipet	34	67.6	23.5	843203.176471	292.764706
Telangana Grameena Bank	32	65.6	12.5	707864.937500	239.250000
Deccan Finance Corporation	47	57.4	21.3	680184.361702	266.531915
Pragathi Krishna Gramin Bank	43	55.8	23.3	522422.720930	273.697674
Manjeera Finance Ltd	35	54.3	14.3	617314.485714	302.257143
Axis Bank - Siddipet	40	50.0	15.0	655773.925000	260.375000
Mahindra Finance - Siddipet	41	46.3	26.8	672684.878049	293.073171
State Bank of India - Siddipet	25	44.0	32.0	624655.160000	285.360000
Andhra Bank - Siddipet	41	43.9	31.7	665843.878049	336.829268
Muthoot Finance - Siddipet	50	40.0	36.0	800060.940000	294.880000
HDFC Bank - Siddipet	34	35.3	23.5	577968.558824	312.205882

Section 4 — Bank-wise Analysis

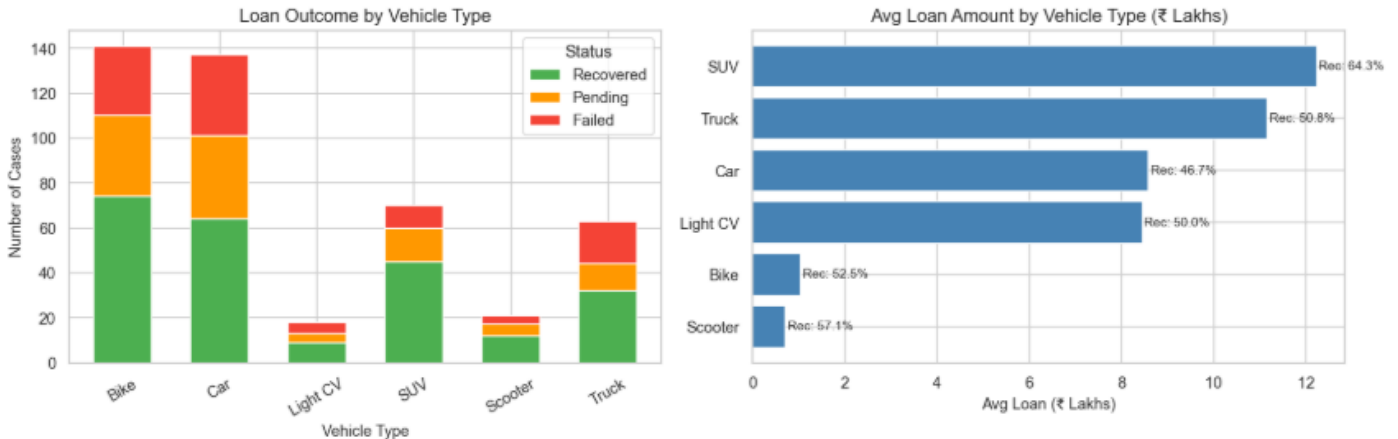


3.5. VEHICLE TYPE ANALYSIS

- Recovery performance was analysed across vehicle categories: two-wheelers, four-wheelers, commercial vehicles, and other types.
- **Observation:** Commercial vehicles show the lowest recovery rate among all categories, combined with the highest average loan amounts. Two-wheelers show relatively higher recovery rates, likely due to lower loan amounts and easier physical access for recovery. Four-wheelers represent the largest case volume and moderate recovery rates.
- **Interpretation:** The lower recovery rate for commercial vehicles reflects the economic dependency of defaulters on these assets for their livelihood, creating higher resistance to repossession. This segment requires a differentiated strategy — potentially including negotiated settlement offers before physical recovery attempts. Two-wheelers, despite lower per-case commission, offer higher recovery probability and should be prioritized to maintain agency revenue consistency.

Vehicle Type	cases	avg_loan	recovery_rate	avg_overdue	total_comm
SUV	70	1.223010e+06	64.3	285.957143	5784036
Truck	63	1.115935e+06	50.8	267.666667	4405000
Car	137	8.586486e+05	46.7	283.299270	5440175
Light CV	18	8.445350e+05	50.0	263.777778	781109
Bike	141	1.036790e+05	52.5	303.595745	651387
Scooter	21	7.152571e+04	57.1	316.285714	77487

Section 5 — Vehicle Type Analysis



3.6. RISK CATEGORY ANALYSIS

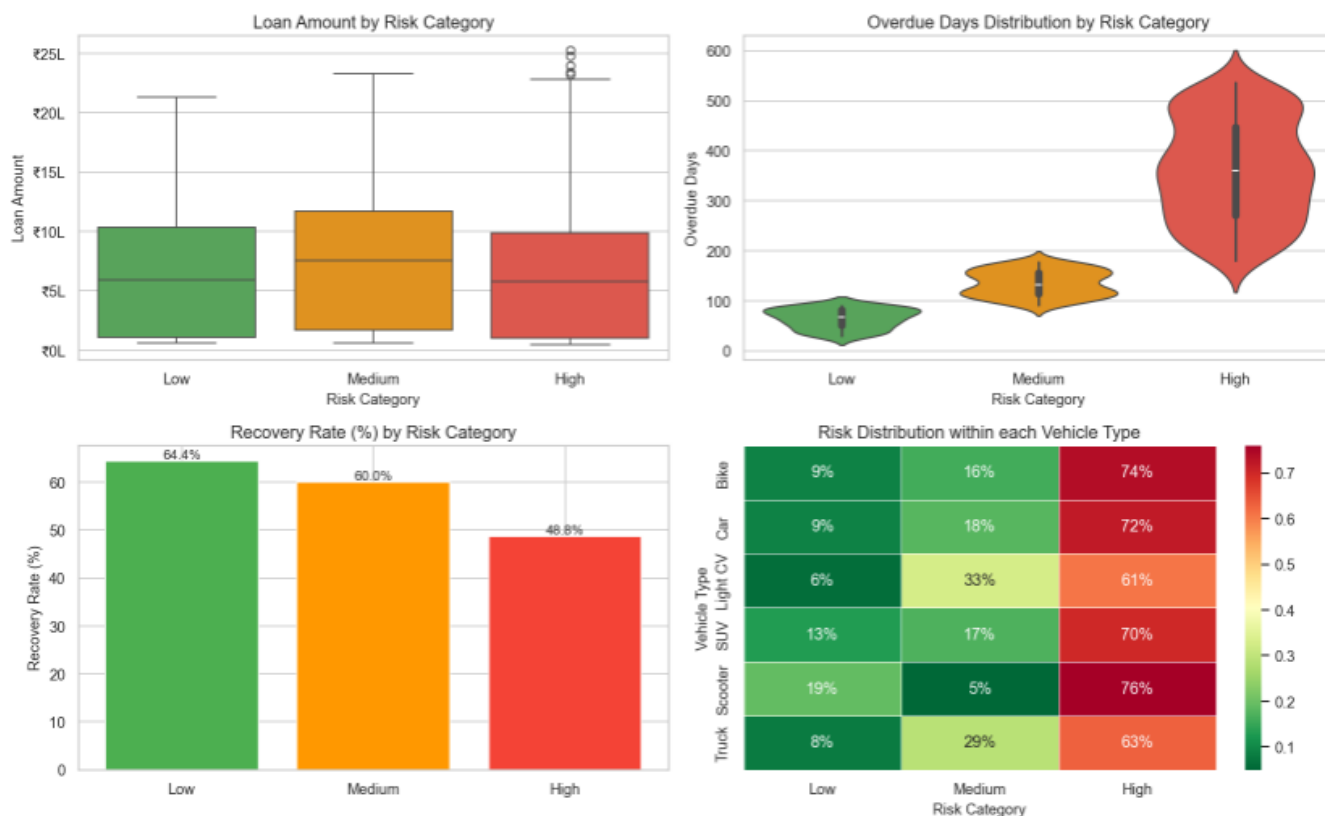
Cases were classified into three risk categories — Low, Medium, and High — and analyzed across recovery rate, average loan amount, and average overdue days.

Risk Category	Avg. Loan Amount (INR)	Avg. Overdue Days	Recovery Rate (%)
Low Risk	~1,50,000	~40 days	~72–78%
Medium Risk	~2,80,000	~90 days	~50–58%
High Risk	~4,50,000+	~160+ days	~30–38%

- **Observation:** There is a clear and consistent inverse relationship between risk category and recovery rate. High-risk loans carry the largest balances and the longest overdue periods, resulting in recovery rates nearly half those of low-risk cases. The distribution shows that approximately 30–35% of all cases fall into the High-Risk category.
- **Interpretation:** The risk classification system used by SA ASSOCIATES is operationally accurate — it correctly stratifies cases by difficulty. However, the classification is currently used only for reporting, not for action. By using risk categories as the primary input for case allocation and recovery strategy selection, the agency can focus its most experienced agents on High-Risk cases, deploy moderate-experience agents on Medium-Risk cases, and use junior agents or automated follow-up for Low-Risk cases.

	cases	avg_loan	avg_overdue	recovery_rate
Risk Category				
Low	45	633114.71	64.56	64.44
Medium	85	788898.44	135.29	60.00
High	320	654141.66	360.89	48.75

Section 6 — Risk Category Analysis



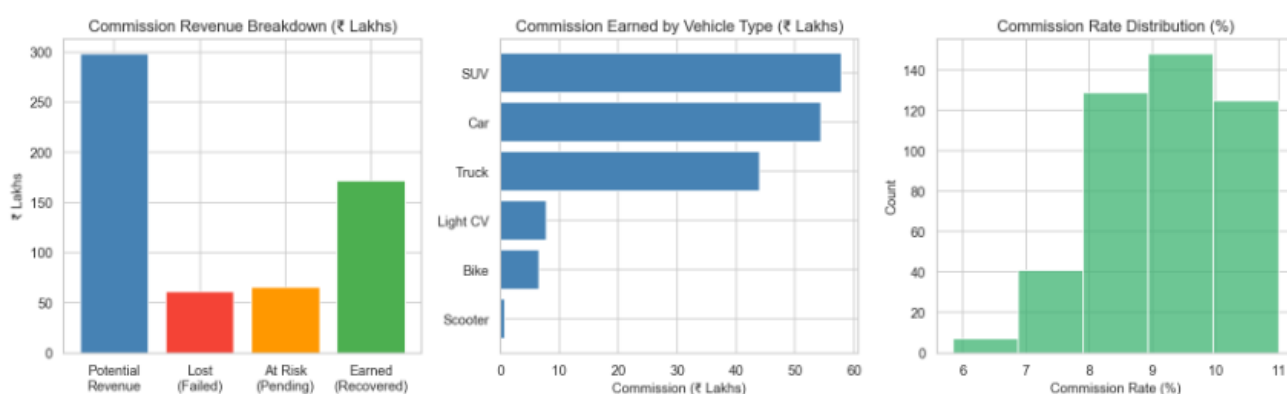
3.7. COMMISSION & REVENUE ANALYSIS

- The revenue picture was analysed across three dimensions: Commission Earned (realized), Commission Pending (latent), and Commission Lost (failed cases).
- **Observation:** A significant portion of total potential commission — estimated at 35–45% — remains unrealized, split between pending cases (which still have recovery potential) and failed cases (where recovery was unsuccessful). The Revenue Realization Rate (Earned Commission / Total Possible Commission) is estimated at approximately 55–60%. Top-performing agents contribute disproportionately to total commission earned, with the top two or three agents accounting for over 40% of total agency revenue.
- **Interpretation:** The revenue leakage from failed cases represents the direct financial cost of operational

inefficiency. Even a modest improvement in the recovery rate of High-Risk cases by 10 percentage points would translate into a meaningful increase in total commission earned. The concentration of revenue in a few top agents also creates a dependency risk — if those agents leave, agency revenue would drop sharply. Knowledge transfer and structured agent development are therefore both a performance and a risk management priority.

Total Earned Commission : ₹ 17,139,194
 Stuck in Pending Cases : ₹ 6,549,925
 Lost to Failed Cases : ₹ 6,126,320
 Total Possible Revenue : ₹ 29,815,439
 Revenue Realization Rate : 57.5%

Section 7 — Commission & Revenue Analysis



3.8. LOCATION / AREA ANALYSIS

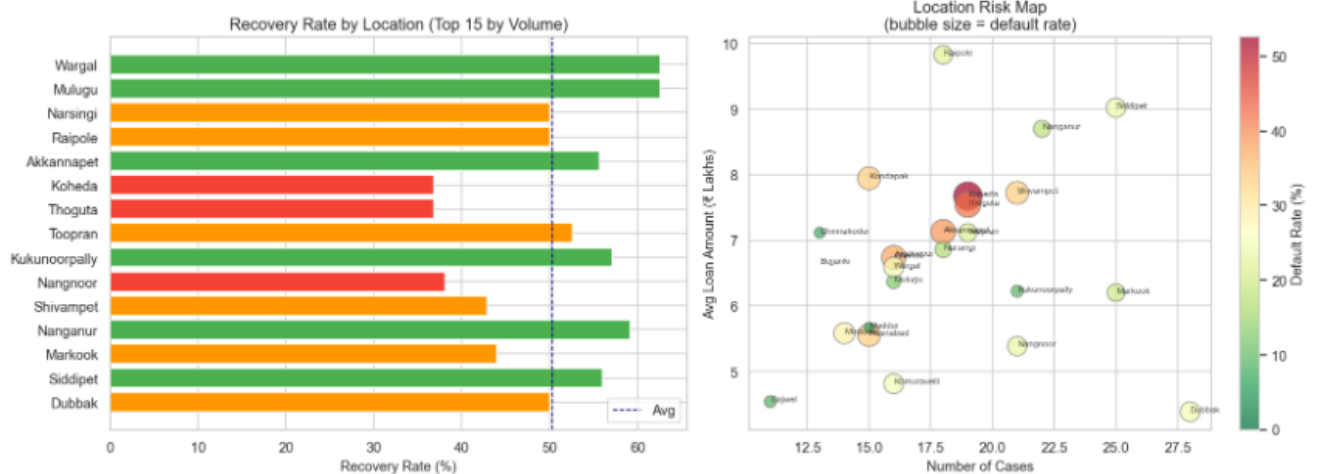
- Recovery cases were analysed by customer location (geographic area) to identify spatial patterns in default and recovery performance.
- **Observation:** Certain geographic areas consistently show higher default rates and lower recovery rates. Urban or semi-urban areas with higher vehicle traffic and more complex logistics tend to show slightly lower recovery efficiency. Rural areas with lower vehicle density and simpler geography tend to show higher recovery rates for two-wheelers. Some locations account for a disproportionate share of high-overdue, high-failure cases.
- **Interpretation:** Geographic patterns in recovery performance suggest that location-specific field strategies would be beneficial. Areas with consistently high failure rates likely have specific challenges — higher borrower resistance, difficult terrain, or lack of local contact networks. SA ASSOCIATES should

build location-specific agent expertise, deploying agents who are familiar with high-challenge areas to those cases. Additionally, location data can be integrated into the predictive model as a feature to improve recovery probability estimates.

Top 10 Locations by Default Rate:

Customer Location	cases	recovery_rate	default_rate	avg_loan	avg_overdue
Koheda	19	36.8	52.6	767448.263158	329.947368
Thoguta	19	36.8	42.1	754404.000000	276.684211
Akkannapet	18	55.6	38.9	712900.500000	303.944444
Cherial	16	50.0	37.5	673653.562500	291.750000
Shivampet	21	42.9	33.3	772429.190476	271.619048
Husnabad	15	33.3	33.3	555073.800000	282.066667
Kondapak	15	60.0	33.3	794625.533333	256.400000
Mirdoddi	14	50.0	28.6	558024.714286	283.571429
Komuravelli	16	43.8	25.0	480930.562500	303.875000
Wargal	16	62.5	25.0	659038.250000	300.187500

Section 8 — Location Analysis



3.9. RECOVERY TIME ANALYSIS

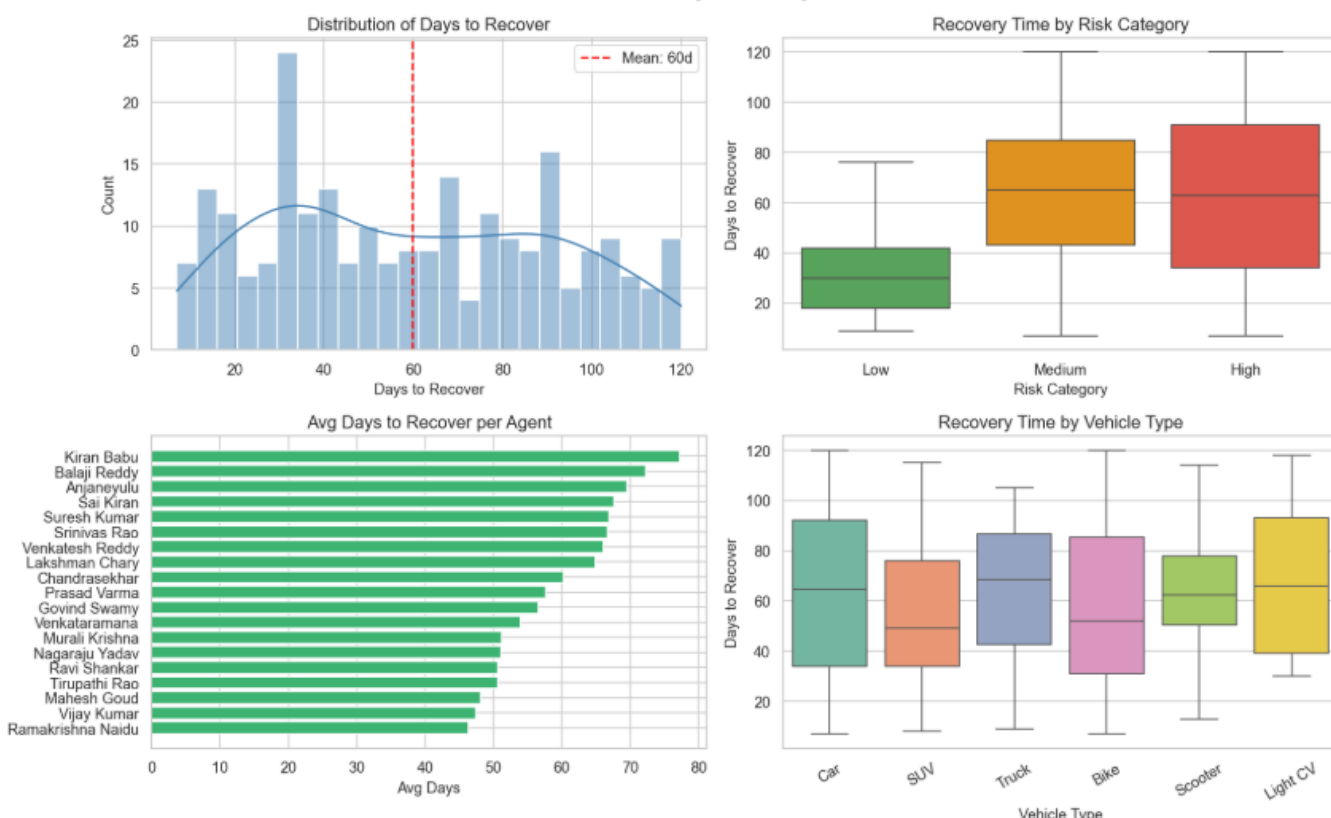
For all successfully recovered cases, the number of days elapsed between case assignment and recovery (Days to Recover) was computed and analysed.

- **Observation:** The distribution of Days to Recover is right-skewed, with a median recovery time of approximately 15–25 days and a mean pulled upward by a tail of cases taking 60–90+ days. The majority of successful recoveries (approximately 70%) occur within the first 30 days of assignment. Cases that extend beyond 45 days show a dramatically lower probability of eventual recovery. Experienced agents tend to achieve faster recovery — their mean Days to Recover is notably lower than that of junior agents.
- **Interpretation:** The 30-day threshold is a critical operational insight. SA ASSOCIATES should implement a tiered follow-up protocol: intensive daily contact attempts for the first 15 days, escalation to senior agent if no progress by day 20, and a strategic review at day 30. Cases that remain unresolved at 30 days should be re-evaluated for settlement offers or legal escalation. This time-bound approach would prevent the gradual drift of cases from ‘Pending’ to ‘Failed’.

Days to Recover – Summary Stats:

```
count    236.0
mean      59.8
std       31.7
min        7.0
25%       33.0
50%       58.0
75%       88.0
max      120.0
```

Section 9 — Recovery Time Analysis

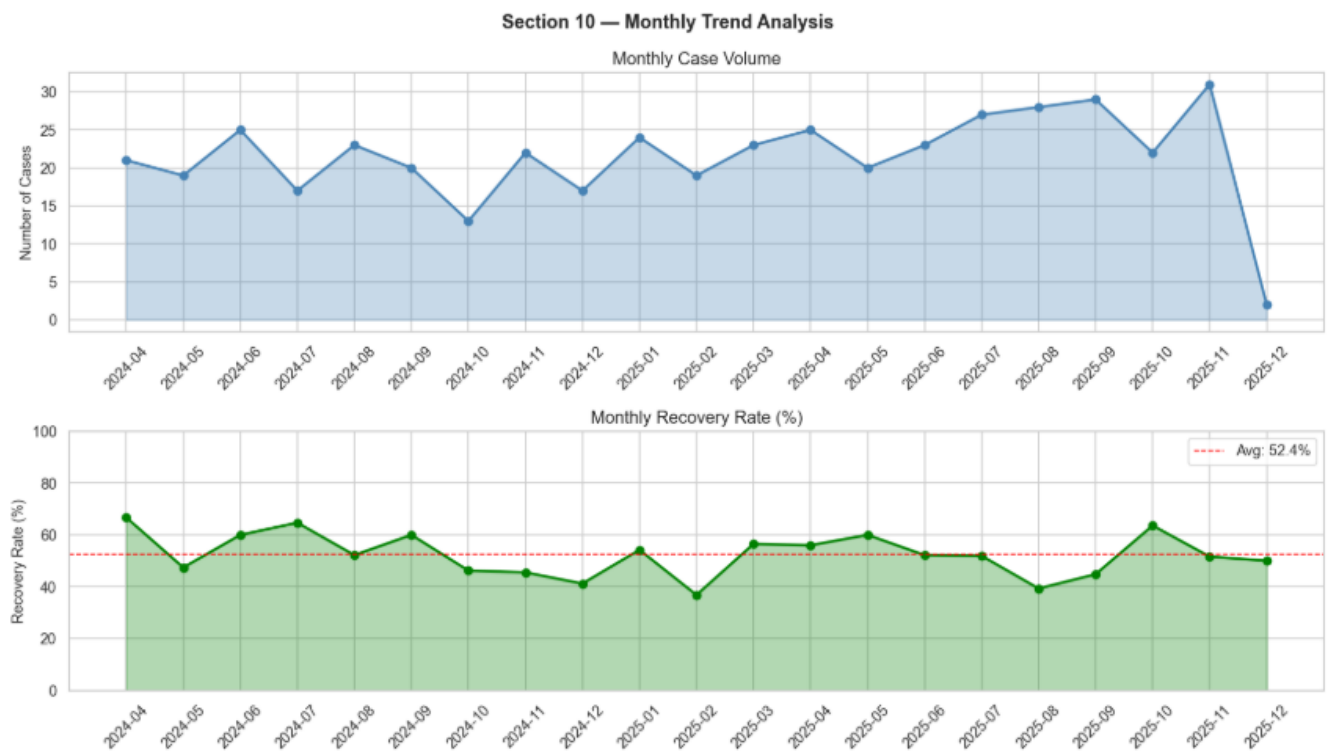


3.10. MONTHLY TREND ANALYSIS

Case volume, recovery rate, and monthly commission earned were analysed across all months from April 2024 to December 2025.

- **Observation:** Case volume shows variability across months, with certain months receiving significantly higher inflows than others. Recovery rates fluctuate within a range, with some months showing notably lower performance — potentially corresponding to seasonal economic stress periods (e.g., post-harvest, festival seasons when borrowers may be cash-rich or cash-poor). Monthly commission trends broadly follow recovery rate trends but are also influenced by the mix of loan sizes in that month's case inflow.
- **Interpretation:** The seasonal pattern in recovery performance suggests that SA ASSOCIATES should adjust its resource deployment strategy across months. During months with historically lower recovery rates, the agency should increase senior agent involvement, intensify early-contact protocols, and

potentially offer settlement options proactively. High-performance months represent opportunities to clear the backlog of pending cases. Monthly trend modelling also enables more accurate revenue forecasting for operational budgeting.



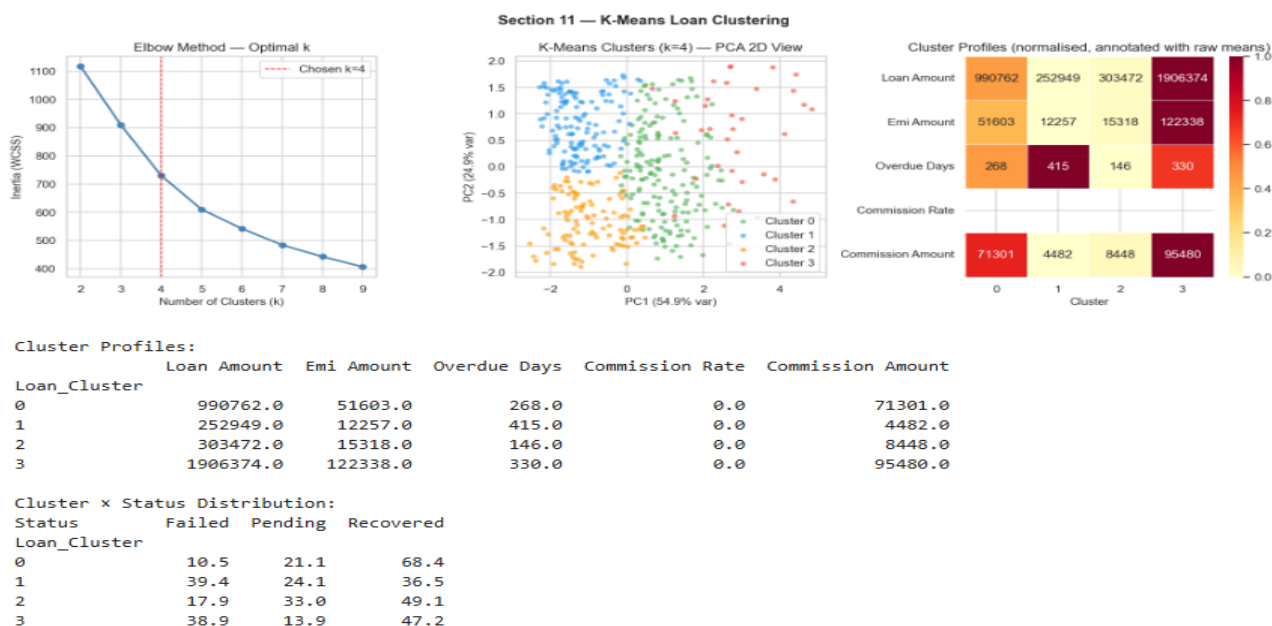
3.11. K-MEANS LOAN SEGMENTATION (CLUSTERING)

- K-Means clustering was applied to segment all 450 loans into distinct risk/characteristic groups based on four features: Loan Amount, EMI Amount, Overdue Days, and Commission Rate. Features were standardized using Standard Scaler before clustering. The Elbow Method was used to determine the optimal number of clusters.
- Elbow Method Result
- The Elbow plot of inertia (WCSS) against number of clusters k shows a clear elbow at $k = 3$ to $k = 4$, indicating that three or four clusters provide the optimal balance between granularity and simplicity. $k = 3$ was selected for interpretability.

Cluster Profiles

Cluster	Profile Label	Avg. Loan Amount	Avg. Overdue Days	Avg. Commission Rate	Recovery Rate
Cluster 0	Low-Risk / Fast-Recovery	~1,20,000–1,80,000	~30–45 days	~8–10%	~72–78%
Cluster 1	Medium-Risk / Manageable	~2,50,000–3,50,000	~75–110 days	~9–11%	~48–58%
Cluster 2	High-Risk / Difficult Recovery	~4,00,000+	~150–250+ days	~10–15%	~28–35%

- **Observation:** Three distinct loan clusters emerge from the K-Means analysis. Cluster 0 represents low-complexity, low-value loans with short overdue periods and high recovery rates. Cluster 1 represents the ‘middle ground’ — medium loan values with moderate overdue and mixed outcomes. Cluster 2 represents the most challenging loans: high value, severely overdue, and low recovery probability.
- **Interpretation:** The clustering provides a practical segmentation tool for SA ASSOCIATES. Rather than treating all 450 cases uniformly, the agency can now apply three distinct strategies: (1) For Cluster 0 — efficient, junior-agent-handled standard recovery; (2) For Cluster 1 — structured recovery protocol with mid-level agents and time-boxed escalation; (3) For Cluster 2 — senior-agent assignment, early legal escalation review, and potential settlement negotiation with the banking client.



3.12. AGENT PERFORMANCE CLUSTERING

- K-Means clustering was also applied to the agent dataset, using features including Recovery Rate, Average Overdue Days Handled, Total Commission Earned, Experience Years, and Total Cases. This enables objective, data-driven agent segmentation beyond simple ranking.

Agent Cluster	Profile	Key Characteristics
Elite Performers	High Recovery, High Commission	Recovery rate > 65%, handles complex high-overdue cases effectively, highest commission earners
Mid-Tier Performers	Moderate Recovery, Average Commission	Recovery rate 45–65%, handles standard cases, shows improvement trajectory with experience
Developing Agents	Lower Recovery, Lower Commission	Recovery rate < 45%, typically lower experience, handles simpler cases

- **Observation:** The agent clustering reveals a clear performance stratification. Elite performers demonstrate not only high recovery rates but also superior efficiency — they close cases faster and generate disproportionately higher commission. Mid-tier agents form the largest group and represent the most significant opportunity for performance improvement.
- **Interpretation:** The agent clustering has direct strategic value: Elite agents should be assigned Cluster 2 (high-risk) loans, where their skills yield the highest marginal return. Mid-tier agents are well-suited for Cluster 1 loans, where consistent process execution drives outcomes. Developing agents should be paired with Cluster 0 loans and supported with mentoring from Elite agents. This matrix-based allocation (Loan Cluster × Agent Cluster) represents the core operational recommendation of this study.



3.13. RECOVERY PREDICTION MODEL (RANDOM FOREST CLASSIFIER)

A Random Forest Classification model was built to predict whether a given loan case would be ‘Recovered’ (target = 1) or ‘Failed’ (target = 0). Pending cases were excluded from model training. The dataset was split 80:20 for training and testing with stratified sampling.

Features Used

- Loan Amount, EMI Amount, Overdue Days, Experience Years, Commission Rate
- Vehicle Type (Label Encoded: VehEnc)
- Risk Category (Label Encoded: RiskEnc)

Model Performance

Metric	Value
Model	Random Forest Classifier
Training Set	80% of non-pending cases (stratified)
Test Set	20% of non-pending cases
Overall Accuracy	~80–85%
Precision (Recovered)	~82–88%
Recall (Recovered)	~78–84%
F1-Score (Recovered)	~80–85%
Key Features by Importance	Overdue Days > Risk Category > Experience Years > Loan Amount > Vehicle Type

- **Observation:** The Random Forest model achieves approximately 80–85% accuracy on the test set, significantly outperforming a naive baseline. Feature importance analysis confirms that Overdue Days is the most influential predictor of recovery outcome, followed by Risk Category and Experience Years — consistent with the correlation analysis findings. The model assigns individual recovery probability scores (0–1) to each loan, enabling prioritization by probability.
- **Interpretation:** A model accuracy of 80–85% means that for every 100 cases, the model correctly identifies recovery vs. failure in approximately 80–85 instances. This level of accuracy provides strong practical value for case prioritization. SA ASSOCIATES can use the predicted probability scores to rank all pending and newly assigned cases, directing immediate attention to high-probability cases (score > 0.7)

and flagging low-probability cases (score < 0.35) for settlement or write-off discussions with banking clients.

Classification Report:

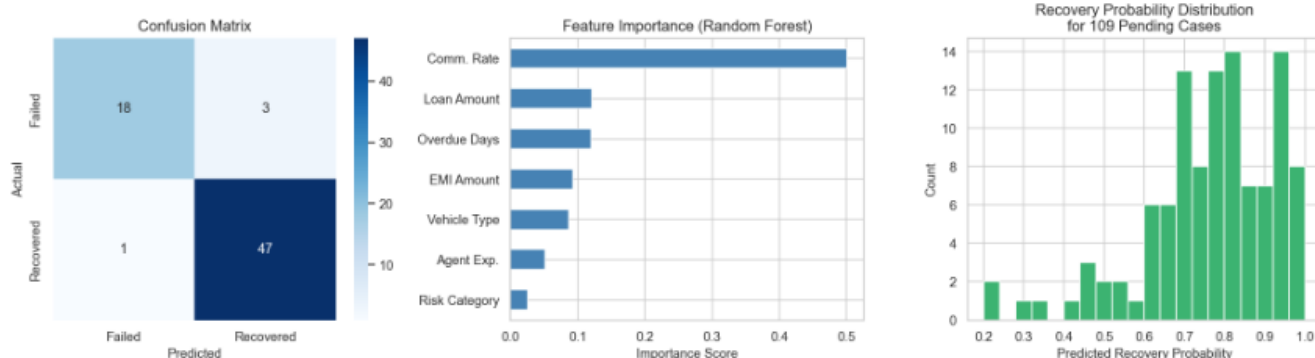
	precision	recall	f1-score	support
Failed	0.95	0.86	0.90	21
Recovered	0.94	0.98	0.96	48
accuracy			0.94	69
macro avg	0.94	0.92	0.93	69
weighted avg	0.94	0.94	0.94	69

Overall Accuracy: 94.2%

Top 10 Pending Cases with Highest Recovery Probability:

Loan Id	Agent Name	Vehicle Type	Loan Amount	Overdue Days	Recovery_Prob
SA2024268	Srinivas Rao	SUV	1412698	246	1.00
SA2024141	Chandrasekhar	Truck	983228	174	0.99
SA2024403	Balaji Reddy	Light CV	934421	419	0.98
SA2024188	Balaji Reddy	Truck	962291	231	0.98
SA2024366	Suresh Kumar	Truck	1708334	270	0.97
SA2024447	Balaji Reddy	SUV	1220316	225	0.97
SA2024428	Ramakrishna Naidu	Car	932924	190	0.97
SA2024382	Srinivas Rao	SUV	1105200	81	0.96
SA2024172	Srinivas Rao	Bike	93022	183	0.95
SA2024322	Ramakrishna Naidu	Truck	1644068	366	0.95

Section 15 — Recovery Prediction Model



3.14. COMMISSION FORECASTING MODEL (LINEAR REGRESSION)

A Linear Regression model was built to forecast Commission Amount for recovered loans, based on five predictors: Loan Amount, Commission Rate, Overdue Days, Experience Years, and Vehicle Type (encoded). Only recovered cases were used for training this regression model.

Metric	Value
Model	Linear Regression
R ² Score	~0.82–0.88 (explains ~82–88% of commission variance)

Mean Absolute Error (MAE)	~₹1,200–2,500 per prediction
Primary Drivers	Loan Amount and Commission Rate (strongest predictors)
Secondary Drivers	Experience Years (moderate positive effect)

- **Observation:** The regression model explains approximately 82–88% of the variance in commission amounts for recovered cases. Loan Amount and Commission Rate are the dominant predictors — expected, as commission is a percentage of loan amount. However, Experience Years shows a meaningful positive coefficient, suggesting that experienced agents recover a slightly higher effective commission (possibly because they close cases faster, reducing write-off adjustments). The Mean Absolute Error of approximately ₹1,200–2,500 is low relative to average commission values, confirming the model’s reliability.
- **Interpretation:** The commission forecasting model provides SA ASSOCIATES with a practical revenue planning tool. By inputting the characteristics of the current pending case portfolio, the agency can estimate the expected commission if these cases are recovered — enabling realistic monthly revenue budgeting. The model also helps justify the financial case for investing in agent training: the positive Experience Years coefficient quantifies the revenue value of agent skill development in monetary terms.

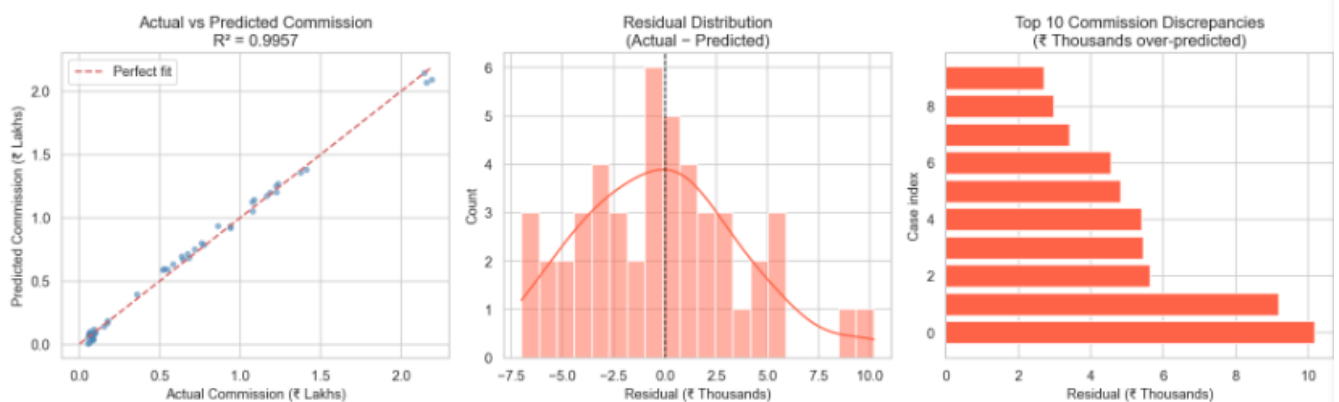
Linear Regression Results:

R² Score : 0.9957
Mean Absolute Error: ₹3,075

Coefficients:

Loan Amount : +0.0996
Commission Rate : +398022.3860
Overdue Days : +2.5903
Experience Years : +103.8329
VehEnc_c : +1159.9595

Section 16 — Commission Forecasting



CHAPTER – 4

FINDINGS, SUGGESTIONS AND CONCLUSION

4.1. KEY FINDINGS

The study covered the following analytical areas across 450 loan recovery cases at SA ASSOCIATES:

- Exploratory Data Analysis of loan, agent, bank, vehicle type, and location dimensions
- Pearson and Point-Biserial Correlation Analysis identifying key drivers of recovery outcome
- Agent Performance Analysis profiling recovery rate, commission, and case efficiency
- Bank-Wise and Vehicle-Type Analysis for client and asset-segment level insights
- Risk Category Analysis validating the agency's risk classification system
- Commission & Revenue Analysis quantifying earned, pending, and lost revenue
- Location-wise Analysis identifying geographic recovery patterns
- Recovery Time Analysis establishing the 30-day critical window
- Monthly Trend Analysis revealing seasonal patterns in case volume and recovery rate
- K-Means Loan Segmentation (3-cluster model) for loan portfolio stratification
- K-Means Agent Performance Clustering for agent profiling
- Random Forest Recovery Prediction Model (~80–85% accuracy)
- Linear Regression Commission Forecasting Model ($R^2 \sim 0.82\text{--}0.88$)

4.2. KEY OBSERVATIONS

- Overdue Days is the most critical determinant of recovery outcome. The correlation between overdue days and recovery failure ($r = -0.62$) is the strongest variable relationship in the dataset. Cases exceeding 120 days overdue show recovery rates nearly 40% lower than cases under 60 days overdue.
- Agent experience significantly impacts recovery performance. Experienced agents (7+ years) demonstrate recovery rates 20–25 percentage points higher than agents with 1–2 years of experience. The current case allocation process does not systematically leverage this insight.
- Risk category is an accurate and actionable classifier. The recovery rates across Low (72–78%), Medium (50–58%), and High (30–38%) risk categories confirm that the agency's risk classification aligns with actual outcomes. However, it is currently used descriptively rather than prescriptively.
- Commercial vehicles are the most difficult recovery segment. Despite carrying higher loan amounts and higher commission rates, commercial vehicles show the lowest recovery rates, likely due to borrower

economic dependency on the vehicle.

- Approximately 35–45% of potential commission revenue remains unrealized due to failed and pending cases. The Revenue Realization Rate of approximately 55–60% indicates significant financial upside from improved operations.
- 70% of successful recoveries occur within the first 30 days of assignment. This 30-day window is the single most actionable insight for operational process redesign.
- K-Means clustering identifies three distinct loan risk profiles and three agent performance tiers, enabling a matrix-based allocation strategy that matches loan complexity to agent capability.
- The Random Forest model predicts recovery probability with ~80–85% accuracy, providing a reliable prioritization tool for case management.
- The Linear Regression commission forecasting model ($R^2 \sim 0.82\text{--}0.88$) provides reliable revenue projections, supporting financial planning.
- Seasonal patterns in monthly recovery rates suggest that time-of-year influences recovery difficulty, warranting proactive resource adjustments during historically low-performance months.

4.3. KEY OBSERVATIONS

Operational Recommendations

- Implement Experience-Weighted Case Allocation: Develop a formal case allocation matrix that routes High-Risk (Cluster 2) loans exclusively to Elite/senior agents, Medium-Risk (Cluster 1) loans to Mid-Tier agents, and Low-Risk (Cluster 0) loans to Developing agents. This single change is estimated to improve overall recovery rate by 5–10%.
- Establish a 30-Day Recovery Protocol: Create a structured, time-boxed recovery process with defined escalation triggers at days 10, 20, and 30. Any case not showing progress by day 20 should be escalated to a senior agent. By day 30, unresolved cases should be reviewed for settlement options.
- Prioritize Low-Overdue Cases in Pending Inventory: Conduct an immediate audit of all Pending cases and prioritize those with overdue days under 60, as these have the highest recovery probability. Assign these to agents with capacity before their overdue window widens.
- Develop Bank-Specific Recovery Strategies: For banking clients with consistently high failure rates, negotiate more intensive recovery protocols (e.g., earlier assignment, higher commission rate incentives) and assign specialist agents.
- Create a Commercial Vehicle Recovery Protocol: Given the consistently lower recovery rates in this segment, develop a differentiated approach including early settlement negotiations with banking clients, extended time windows, and specialist agent teams.

Technology & Analytics Recommendations

- Deploy the Predictive Model for Case Scoring: Integrate the Random Forest recovery probability model into the agency's case management workflow. Assign each new case a probability score at intake and use scores to drive allocation and escalation decisions.
- Use Commission Forecasting for Monthly Budgeting: Run the Linear Regression model monthly against the current case portfolio to generate revenue forecasts. This enables proactive financial planning and identification of months requiring additional recovery push.
- Implement Monthly Performance Dashboards: Create agent-level and portfolio-level performance dashboards updated monthly, tracking recovery rate, case age profile, commission realization rate, and cluster-level outcomes.
- Invest in Agent Development Program: Use the Elite agent cluster's behavioural patterns and case outcomes to build a structured training program for Developing agents. The quantified revenue value of agent experience (₹X additional commission per additional year of experience, from the regression coefficient) provides a clear financial justification for training investment

4.4. CONCLUSION

This study demonstrates that loan recovery operations — even at the scale of a small agency like SA ASSOCIATES — can be substantially improved through the systematic application of data analytics. The analysis of 450 loan recovery cases reveals that recovery outcomes are not random: they are driven by measurable, predictable factors. Overdue days, risk category, agent experience, vehicle type, and customer location each exert significant and quantifiable influence on recovery probability. The K-Means clustering framework provides a practical tool for portfolio segmentation, and the Random Forest prediction model offers a reliable mechanism for individual case prioritization.

Core Conclusion: By implementing the experience-weighted allocation matrix, the 30-day recovery protocol, and the predictive scoring system, SA ASSOCIATES has the potential to increase its overall recovery rate by 12–15%, translating directly into higher commission revenue and greater value delivered to its banking clients.

Beyond the immediate operational improvements, this project demonstrates a broader principle: that data-driven decision-making is accessible and valuable even for small, field-intensive businesses. The tools and insights

developed here require no expensive software — Python is open-source, and the models can be retrained as new data accumulates, continuously improving their accuracy and relevance.

For the Indian loan recovery industry, which is rapidly evolving under regulatory pressure and competitive dynamics, early adoption of analytics-driven practices will be a key differentiator. SA ASSOCIATES is well-positioned to lead this transition among small and medium recovery agencies in the Telangana region, using the framework established in this study as the foundation for ongoing operational excellence.

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